

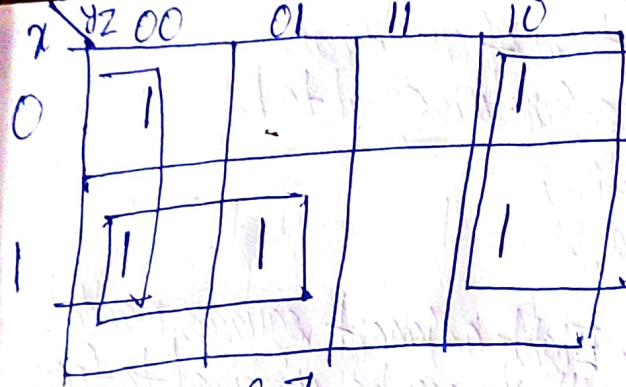


Figure 3.7

The simplified function is: $\boxed{z' + xy'}$

3.3 Four-variable Map

m_0	m_1	m_3	m_2
m_4	m_5	m_7	m_6
m_{12}	m_{13}	m_{15}	m_{14}
m_8	m_9	m_{11}	m_{10}

(a)

	wz	00	01	11	10
w	00	$w'x'y'z'$	$w'x'y'z$	$w'x'yz$	$w'x'yz'$
	01	$w'xy'z'$	$w'xy'z$	$w'xyz$	$w'xyz'$
	11	$wxy'z'$	$wxy'z$	$wxyz$	$wxyz'$
	10	$wxy'z'$	$wxy'z$	$wxyz$	$wxyz'$
w					

(b)

Figure 3.8

- In (a) are listed the 16 minterms and the squares assigned to each.
- In (b) the map is redrawn to show the relationship with the four variables.
- The rows and columns are numbered in a reflected-code sequence, with only one digit changing value between two adjacent rows or columns.
- The minterm corresponding to each square can be obtained from the concatenation of the row number with the column number. For example, the numbers of the third row (11) and the second column (01), when concatenated, give the binary number 1101, the binary equivalent of decimal 13. Thus the square in the third row and second column represents minterm m_{13} .
- The combination of adjacent squares that is useful during the simplification process is easily determined from the inspection of the four-variable map:
 - 1) One square represents one minterm giving a term of four literals.
 - 2) Two adjacent squares represent a term of three literals.
 - 3) Four adjacent squares represent a term of two literals.